Navigating the Sustainability Disclosure Ecosystem

A diagnostic framework for European seafood processing SMEs

Moving from reactive compliance to proactive sustainability strategy

Introduction and Methodology — Working Paper

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1. Introduction

1.1 Sustainability as strategy

Companies are increasingly faced with sustainability demands originating from regulatory, market, and societal stakeholders. The way firms respond to these demands varies along a continuum of sustainability maturity, between two ideal-typical postures (Russo & Fouts, 1997; Sharma & Vredenburg, 1998).

A reactive approach implies that the firm meets the bare minimum of stakeholder demands by acting only when external pressure arises. Sustainable practices are not specifically developed and remain peripheral to the business model. Firms with a reactive posture comply with the minimum sustainability prerequisites required to operate but are unlikely to benefit from performance gains associated with deeper sustainability integration. A proactive approach, in contrast, implies voluntary engagement that exceeds the legal minimum, contributing to environmental and social development as well as long-term economic performance. Empirical evidence from CSR research in small and medium-sized enterprises (SMEs) suggests that proactive economic-related CSR generates positive direct effects on financial performance over the long term (Torugsa et al., 2013).

The barrier to proactive engagement, particularly for resource-constrained firms, is not lack of intent but external-input overload. Fragmented and complex sustainability governance pushes SMEs toward reactive compliance behaviour rather than strategic integration (Chan et al., 2022). Where multiple stakeholders demand different information through different channels, the practical effect on SMEs with limited compliance capacity is paralysis or minimum-viable response.

1.2 SME sustainability constraints

SMEs face a well-documented set of constraints that shape their sustainability strategy choices. These include: limited financial and managerial resources; lack of dedicated internal expertise in environmental and social reporting; weak data systems and IT infrastructure; high dependence on a small number of B2B customers; and uncertainty regarding the evolving regulatory landscape (Battisti & Perry, 2011).

In practice these constraints mean that an SME confronted by multiple, overlapping sustainability requirements cannot simply expand its compliance function proportionally. Each new disclosure request from a customer, certifier, investor, or regulator imposes a fixed overhead disproportionate to the firm's size. The resulting incentive is to treat sustainability disclosure as transactional compliance, answering questionnaires as they arrive, rather than as a strategic investment in capabilities that could serve multiple demands simultaneously (Interview).

1.3 Fragmented governance in the seafood sector

Seafood is the most highly traded food product globally, with per capita consumption increasing more rapidly than other animal proteins (Barklay and Miller., 2018). The sector is structurally complex, involving global sourcing networks, multiple processing stages, and exposure to a wide range of sustainability risks including greenhouse gas emissions, marine ecosystem impacts, antimicrobial use, forced labour in fishing fleets, and food safety incidents (Barklay and Miller., 2018).

European seafood processors face this complexity from a position of compounded pressure. In addition to horizontal European legislation addressing waste and packaging (Regulation (EU) 2025/40), sustainability claims (the proposed Empowering Consumers Directive, Directive (EU) 2024/825), and corporate sustainability disclosure (CSRD/ESRS, CSDDD), the sector is subject to a dense overlay of sector-specific regulation: the EU Common Fisheries Policy (Regulation (EU) No 1380/2013), the IUU Regulation (Council Regulation 1005/2008), the recast Control Regulation (2023/2842), the EU Deforestation Regulation (2023/1115, relevant for aquaculture feed sourcing), and the Farm to Fork Strategy under the European Green Deal.

Layered on top of this regulatory architecture is a fragmented landscape of voluntary instruments: certifications (MSC for wild-capture, ASC and BAP for aquaculture, GLOBALG.A.P.), food safety standards (IFS Food, BRCGS, FSSC 22000, benchmarked by GFSI), supply-chain transparency platforms (Sedex/SMETA, EcoVadis), investor frameworks (CDP, SBTi, TCFD, IFRS S2, the Coller FAIRR Protein Producer Index), industry initiatives (GSSI, GDST, Move to Minus 15, ISSF for tuna), and management-system standards (ISO 14001, 45001, 50001). Governance is structurally fragmented, with no single coordinating actor and overlapping but non-identical demands from each (Bush & Oosterveer, 2019).

The empirical implication, surfaced in early scoping interviews for this research, is that European seafood SMEs receive sustainability data requests from multiple actors who individually demand reasonable information but collectively impose an unsustainable response burden. The SME response, predictably, is reactive: comply with the loudest demand, defer the rest, do not invest in capability that would serve more demands than the one in front of them.

1.4 The sustainability reporting landscape

The European Union has positioned corporate sustainability reporting as a key lever for the European Green Deal, intending to improve the availability, comparability, and reliability of sustainability information available to investors, regulators, and value-chain partners (European Commission, 2019). Disclosure is conceptualised not only as a transparency mechanism but as a tool that shapes economic behaviour by directing capital toward sustainable activities.

The Corporate Sustainability Reporting Directive (CSRD; Directive (EU) 2022/2464) operationalises this ambition. CSRD is implemented through the European Sustainability Reporting Standards (ESRS; Commission Delegated Regulation (EU) 2023/2772), which define mandatory disclosure requirements across twelve cross-cutting and topical standards covering environmental, social, and governance dimensions. ESRS introduces the double materiality principle, requiring undertakings to report both on how sustainability matters affect their financial performance and on the firm's own impacts on the economy, environment, and people (EFRAG, 2021).

CSRD scope is restricted to large undertakings and listed entities meeting specified employee, turnover, and balance-sheet thresholds. Under the Omnibus simplification proposals (European Commission, 2025), the scope of CSRD has been further restricted, with smaller previously-in-scope firms now exempted. The majority of European seafood processing companies fall below CSRD thresholds and are therefore not directly subject to ESRS reporting obligations.

To address the disclosure needs of out-of-scope SMEs, EFRAG has developed the Voluntary Sustainability Reporting Standard for SMEs (VSME), comprising a basic and a comprehensive module. VSME is designed to be proportionate to SME constraints, reducing the disclosure load from approximately 1,000 ESRS datapoints to approximately 100. Adoption is voluntary, but the framework is intended to provide a recognised reference set that satisfies the most common information requests SMEs receive from value-chain partners and financial institutions.

In practice, however, two structural gaps are observable. First, the leap from no sustainability reporting at all to even the basic VSME module remains significant for many SMEs (interview). Second, the gap between VSME and full ESRS is substantially larger and is precisely the territory in which non-CSRD seafood SMEs operate when responding to ad-hoc customer questionnaires, certification audits, and investor requests. The sustainability reporting frame, in other words, does not cover the operational reality of European seafood SMEs in the post-CSRD environment.

1.5 Research gap and the diagnostic-tool argument

Existing literature addresses several aspects of this problem in isolation. Sustainability reporting maturity models describe staged organizational evolution in reporting practice (Baumgartner et al., 2010; Cöster et al., 2020). Studies of CSRD implementation costs document the burden faced by in-scope firms (EFRAG, 2025). Stakeholder-pressure literature explains why firms respond to particular demands (Mitchell, Agle & Wood, 1997; Suchman, 1995). Institutional-isomorphism theory provides a lens on why disclosure conventions converge despite formal voluntariness (DiMaggio & Powell, 1983). The seafood-specific literature documents the fragmented private-governance landscape (Bush et al., 2017; Tlusty, 2012; Vandergeest, 2018).

What is missing is an integrated empirical analysis that (i) maps the actual cross-framework disclosure demand facing European seafood SMEs, (ii) identifies where multiple stakeholders converge versus where they diverge, and (iii) translates that empirical pattern into a diagnostic tool that lets an SME self-position on the reactive-to-proactive spectrum and identify which capability investments would most efficiently expand its sustainability response.

This thesis addresses that gap. The empirical material is drawn from the disclosure practice of five large European seafood processors that are CSRD-reporting; this cohort defines the realistic upper bound of seafood-sector disclosure depth. Cross-referenced against the broader stakeholder ecosystem of regulators, customers, investors, and industry initiatives, the analysis identifies disclosure topics that face concentrated cross-stakeholder demand. These hotspots define the high-leverage capability investments for SMEs: topics where building reporting capability once satisfies many demands simultaneously, enabling the SME to shift from reactive question-by-question response to proactive capability-driven engagement.

2. Research questions and contribution

2.1 Research questions

The research is organised around three interconnected questions:

1. **RQ1 - Empirical:** What ESRS topical disclosure requirements are actually reported on by mature European seafood processing companies, and what does this empirical cohort reveal about the practical upper bound of sector disclosure depth?
2. **RQ2 - Analytical:** Across the full landscape of regulatory, customer, investor, and industry frameworks active in the European seafood sector, which disclosure topics constitute hotspots of concentrated cross-stakeholder demand?
3. **RQ3 - Prescriptive:** How can a European mid-sized seafood processor diagnose its current sustainability posture (reactive vs proactive) and prioritize capability investments to progress along that spectrum in a way that satisfies the most pressing cross-stakeholder demands?

2.2 Intended contributions

The thesis contributes to three audiences:

- **For seafood SMEs (primary audience):** a diagnostic framework and a stage-of-development roadmap that translates the complexity of the disclosure ecosystem into a sequence of capability investments tailored to the SME's current posture and identified stakeholder pressures.

- **For academic literature:** a first systematic mapping of cross-stakeholder disclosure demand at the sector level using a defined coverage scale. An extension of sustainability-reporting maturity-model literature with a diagnostic-tool operationalization. An empirically grounded application of the reactive/proactive CSR distinction.
- **For EU policy (secondary):** empirical input to the ongoing simplification debate (Omnibus, VSME extension, sector standards postponement). The cross-stakeholder hotspot analysis identifies where the regulatory and voluntary disclosure ecosystem is converging and therefore where coordination or capping of demand would deliver disproportionate burden reduction for SMEs. This aligns with the Competitiveness Compass target of cutting administrative burden by ≥35% for SMEs (European Commission, 2025).

3. Conceptual framework

3.1 The reactive-proactive sustainability spectrum

The reactive-proactive distinction (Sharma & Vredenburg, 1998; Torugsa et al., 2013) provides the central organising frame. The two postures differ along several dimensions:

Dimension	Reactive posture	Proactive posture
Trigger for action	External demand (questionnaire, audit, regulatory deadline)	Strategic anticipation; voluntary commitments
Disclosure approach	Transactional, question-by-question; minimum viable	Capability-driven, integrated; goes beyond legal minimum
Data infrastructure	Ad-hoc, manual, rebuilt per request	Systematic, automated, reusable across demands
Strategic positioning	Compliance cost center	Competitive differentiation; risk-adjusted financial benefit
Stakeholder relationship	Defensive; minimum communication	Proactive engagement; trust-building

(for now made by me based on: Sharma & Vredenburg, 1998; Torugsa et al., 2013)

Empirical evidence in CSR research indicates that the proactive posture is associated with positive long-term financial performance effects (Torugsa et al., 2013). The barrier to adoption among resource-constrained SMEs is not rejection of the proactive posture in principle, but the absence of a tractable pathway from reactive practice to proactive capability given the fragmented and unpredictable nature of external demand (interview – look for source).

3.2 Stakeholder pressure and fragmented governance

Stakeholder-salience theory (Mitchell, Agle & Wood, 1997) explains how firms triage competing claims based on the legitimacy, urgency, and power of each stakeholder. In contexts of single-stakeholder dominance, the response is clear. In contexts of multiple legitimate stakeholders with overlapping but non-identical demands, case in point for the seafood sustainability ecosystem, the absence of demand coordination creates the fragmentation problem: each stakeholder is individually legitimate, but the cumulative response burden is unmanageable.

Institutional theory (DiMaggio & Powell, 1983) further explains why disclosure conventions converge across firms even where formally voluntary: coercive isomorphism (regulatory pressure), mimetic isomorphism (copying perceived leaders), and normative isomorphism (professionalisation of sustainability functions) all push firms toward similar disclosure choices. For SMEs the mimetic mechanism is particularly relevant because they typically lack the in-house expertise to develop independent positions.

The fragmented governance of the seafood sustainability movement, with regulator, brands, retailers, certifiers, NGOs, and investors all exercising parallel and overlapping demands (Bush et al., 2019), represents an extreme case of the multi-stakeholder challenge. This thesis takes as its starting analytical premise that the SME response to such fragmentation is structurally reactive, and that this reactivity is not a failure of SME intent but a rational adaptation to incoherent external demand.

3.3 The bridge: cross-stakeholder demand convergence

The conceptual bridge from reactive to proactive in this fragmented environment lies in identifying disclosure topics where multiple stakeholders converge. Where the same topic is demanded by a customer (e.g., via a chain-of-custody certification), an investor (via a CDP questionnaire), an industry framework (via GRI 13), and a regulator (via ESRS or a sector-specific instrument), the SME that invests in capability for that topic obtains compound returns: a single capability investment satisfies multiple demands.

Hotspots of cross-stakeholder convergence therefore represent the strategic-investment opportunities through which an SME can begin to move from reactive to proactive. Building disclosure capability on a hotspot topic delivers, by definition, leverage: each unit of capability invested generates response capacity across multiple stakeholders. By contrast, building capability on a topic demanded by only one stakeholder type generates narrower returns and entrenches the transactional posture.

This insight motivates the central analytical construct of the thesis: a heatmap of disclosure topics by stakeholder type, populated from a systematic mapping of the seafood sustainability ecosystem. The hotspots identified in this heatmap form the empirical basis for a stages-of-development roadmap and a diagnostic tool through which an SME can position itself, identify its highest-leverage next investment, and progressively expand its sustainability response.

3.4 The seafood sustainability playing field: an integrative framework

Sections 3.1 to 3.3 establish three building blocks: the reactive-proactive continuum along which firms position themselves (3.1); the fragmented, multi-stakeholder structure of sustainability pressure in the seafood sector (3.2); and the principle that cross-stakeholder convergence creates leverage (3.3). This section integrates these building blocks into a single organising framework, the seafood sustainability playing field, which constitutes the central conceptual contribution of this thesis. The framework serves two purposes: it provides a structured map of the disclosure environment in which seafood firms operate, and, through the construct of capability corridors, it identifies where compliance with mandatory requirements can be deliberately extended into proactive sustainability strategy.

The three structuring axes

The playing field organises the disclosure environment along three axes. The first two are structural, in that they define the position of every requirement, while the third is an attributional overlay.

The necessity axis is the framework's spine. It positions each requirement on a spectrum of obligation that runs from a foundational floor up through four tiers. Beneath the four tiers sits Tier 0, food integrity: the non-negotiable food-safety and food-law floor - General Food Law, the hygiene regulations, and the customer-required food-safety certifications IFS Food, BRCGS and FSSC 22000 - that every EU seafood processor clears before any sustainability question arises. Tier 0 is in scope of the firm but is held in a separate band rather than treated as a sustainability requirement, because it is a precondition of operating. Above it, Tier 1, legal license to operate, comprises requirements without which the firm cannot lawfully operate, such as the catch-certificate regime of the EU IUU Regulation, the Fisheries Control Regulation, the EU Deforestation Regulation and packaging law; the Common Fisheries Policy frames this regime but binds Member States rather than the processor directly. Tier 2, market license to operate, comprises requirements that are legally optional but commercially non-negotiable, because no major retail customer will list a supplier without them, such as MSC or ASC chain-of-custody certification for particular products and the SMETA and EcoVadis social-audit memberships. Tier 3, competitive expectation, comprises requirements that are increasingly normalised, such that their absence constitutes a competitive disadvantage, such as VSME reporting and the value-chain disclosures cascaded from customers under CSRD and its ESRS standards. Tier 4, strategic differentiation, comprises genuinely voluntary commitments that signal leadership. Alongside the tiers, a separate enabling layer (Layer 2) holds the methods, management systems and data infrastructure - the GHG Protocol, the ISO management-system standards, life-cycle-assessment methods - whose necessity is derived from the demand they serve rather than being intrinsic, and which therefore sit outside the necessity grid. This axis is grounded in legitimacy theory (Suchman, 1995), whose distinction between pragmatic, moral and cognitive legitimacy corresponds to the progression from legally enforced to normatively expected obligation, and in institutional theory (DiMaggio & Powell, 1983), whose coercive, mimetic and normative isomorphism mechanisms correspond to the lower, middle and upper tiers respectively. The necessity axis is structurally central because it maps directly onto the reactive-proactive continuum of Section 3.1: a reactive firm operates only within the foundation and Tiers 1 and 2, whereas a proactive firm extends its engagement into Tiers 3 and 4.

The domain axis classifies each sustainability requirement by its subject matter, distinguishing two categories: seafood-sector sustainability (the sector-specific environmental and social topics such as bycatch, stock health, aquaculture welfare, and chain-of-custody traceability) and general corporate sustainability (the cross-sector topics such as greenhouse-gas emissions, energy, labour, and governance). Core operational and food-integrity requirements - food safety, hygiene, basic legal compliance - are not treated as a third domain but as the Tier 0 foundation described above, because they are a precondition of operating rather than a position on the sustainability spectrum. The domain axis thus presents the distinction between seafood-specific and general sustainability requirements as two adjacent columns, while food integrity underlies both.

The source axis attributes each requirement to its primary stakeholder driver: regulator, customer, investor, or industry and civil society. Unlike the first two axes, source is not used to structure the framework but is applied as a colour-coded overlay, because stakeholder drivers cross-cut both necessity and domain. Source is assigned by a single test: does the instrument legally bind the processor? If it does, the driver is the regulator (for example the EU Deforestation Regulation, packaging law, or occupational-safety law). If it does not, the driver is the stakeholder that transmits the demand - a customer audit codes to customer, a capital-market expectation to investor - and self-adopted voluntary frameworks are coded by their primary intended audience, with investor-facing standards such as IFRS S1/S2 and TNFD coding to investor and multistakeholder or NGO-led standards such as GRI and SBTi coding to industry and civil society. This axis draws on stakeholder-salience theory (Mitchell, Agle & Wood, 1997): the same requirement may be demanded with differing power, legitimacy and urgency by different actors.

A second overlay, obligation locus, records separately from source whether the processor is itself the bound party. A requirement is direct where the processor is the obligated, certified or reporting entity; indirect where the obligation sits elsewhere in the value chain and reaches the processor through its own disclosure duty or a buyer's demand, in which case it is placed at the tier of that transmission mechanism; and context where the instrument binds a Member State or another actor with no mechanism that transmits an obligation to the processor at all. Source and locus are orthogonal. The EU Deforestation Regulation is regulator-driven and direct; the fisheries-labour Directive is customer-driven, because it binds vessel owners and reaches the processor through social audits, yet its locus is indirect; the voluntary VSME standard is investor-driven and direct; and CSRD reaches the sub-threshold processor as a customer-driven, indirect value-chain demand. Holding the two dimensions apart prevents the common error of reading the legally bound party off the identity of the actor who actually applies the pressure.

Rendering source and locus as overlays rather than axes makes visible a pattern central to the argument of this thesis, namely that necessity, domain, source and locus are correlated but not identical. Tier 1 requirements are predominantly regulator-driven and direct, Tier 2 predominantly customer-driven, and the voluntary upper tiers spread across investor and industry drivers. It is the visible gradient of this correlation, rather than its assumption, that the framework contributes. Figure 1 presents the populated framework, with the necessity tiers as rows, the two domain categories as columns, each instrument positioned in its cell and colour-coded by source, and its obligation locus marked.

Capability corridors and the compliance-to-strategy synergy

A grid of requirements organised by necessity and domain is a descriptive map; it does not, by itself, explain how a firm moves from reactive compliance to proactive strategy. The construct that performs this function is the capability corridor. A capability corridor is an underlying organisational capability, a body of data, a process, or an information system, upon which multiple requirements across different cells of the framework draw. The construct is grounded in the resource-based view of the firm (Barney, 1991) and the dynamic-capabilities literature (Teece, 2007): a capability, once built, is fungible across the demands that require it.

Two corridors illustrate the construct. The traceability corridor runs from a Tier 1 anchor, the catch-certificate and lot-tracking obligations of the IUU and Control Regulations, through Tier 2 chain-of-custody certification (MSC, ASC) and the GDST traceability standard, into Tier 3 value-chain disclosure under VSME and ESRS. The supplier-engagement corridor runs from a Tier 2 anchor, the supplier-control requirements of food safety certifications such as IFS Food and BRCGS, through SMETA social auditing into Tier 3 value-chain worker disclosure. In each case a single capability, built to satisfy a lower-tier mandatory requirement, also supplies a substantial share of the infrastructure required by higher-tier strategic disclosures.

This observation yields the framework's central strategic proposition. Because a capability built for a lower-tier mandatory requirement subsidises the higher-tier requirements that draw on the same corridor, the marginal cost of proactive sustainability disclosure is considerably lower than it appears when each requirement is assessed in isolation. A firm that builds traceability capability because Tier 1 law compels it has, in doing so, already financed a substantial portion of the infrastructure required for MSC certification and for ESRS value-chain disclosure. The proactive sustainability strategy, on this account, is to a large degree the deliberate recognition and extension of capabilities the firm was compelled to build for legal survival. The reactive firm, by contrast, rebuilds capability requirement by requirement, never recognising the corridor, and so perpetually pays the full isolated cost of each demand. The corridors are therefore where compliance becomes strategy, and they constitute the conceptual bridge, anticipated in Section 3.3, between the reactive and proactive postures.

Relationship to the empirical analysis

The playing field is a conceptual framework; the empirical chapters of this thesis populate and test it. Phases 1 and 2 of the methodology (Sections 4.3 and 4.4) identify the disclosure topics and the instruments that occupy the framework's cells. Phase 3 (Section 4.5) codes the demand each instrument places on each topic, producing the cross-framework demand matrix that empirically substantiates the source overlay and reveals where corridors run. Phase 4 (Section 4.6) operates on this matrix to identify cross-stakeholder hotspots and to assign topics to development stages.

A clarification is required regarding the relationship between the framework's necessity tiers and the development stages produced by the Phase 4 analysis, because the two are related but distinct. The necessity tiers are a structural property of each requirement, namely how obligatory it is, and form the rows of the framework. The development stages are an empirical product of the demand analysis, namely how many stakeholder types converge on a topic, and identify where leverage is concentrated. A topic may be high in necessity but narrow in cross-stakeholder demand, or low in necessity but broad in demand. The roadmap that this thesis ultimately proposes therefore sequences capability investment by necessity first and by cross-stakeholder leverage second: a firm addresses its Tier 1 obligations as a precondition of operating, and, among the topics within reach at each tier, prioritises those that sit on high-leverage corridors. Necessity defines what must be done; the demand analysis defines, among what must or could be done, where to begin.

4. Methodology

4.1 Research design overview

The research employs a mixed-methods empirical design organised in five phases:

- **Phase 1 - Universe construction.** Defining the set of disclosure topics in scope for the analysis, drawn from cohort empirical practice (ESRS-derived) and supplemented with sector-material topics outside ESRS (GRI).

- **Phase 2 - Landscape mapping.** Cataloguing the regulatory, customer-driven, investor-driven, and industry-driven frameworks active in the European seafood sustainability ecosystem.
- **Phase 3 - Cross-framework demand mapping.** Coding each disclosure topic against each priority framework on a defined coverage scale, producing the central demand matrix.
- **Phase 4 - Hotspot analysis and stage assignment.** Identifying topics with concentrated cross-stakeholder demand. Assigning topics to development stages; constructing the diagnostic tool.
- **Phase 5 - Case study validation.** Applying the framework to one or two SME case studies to test usability and refine the diagnostic tool.

4.2 Empirical scope: the large-reporter cohort

The empirical foundation is the disclosure practice of five large European seafood processing companies that are CSRD-reporting or have published sustainability reports of comparable depth: Bolton Group, A. Espersen A/S, Mowi ASA, Nomad Foods Limited, and Profand Group. These five represent the realistic upper bound of European seafood sector disclosure depth and span both wild-capture and aquaculture operations, branded and private-label, listed and unlisted.

The data sources are each company's most recent publicly available sustainability report (2023 or 2024 reporting cycle), supplemented where available by their CSRD-aligned ESRS-index appendices. The unit of empirical analysis is the ESRS Disclosure Requirement (DR); each report was coded for whether the company reports against each of the 70 ESRS topical DRs (E1–E5, S1–S4, G1).

4.3 Phase 1 - Universe construction

The disclosure-topic universe used in the cross-framework mapping is constructed in four steps.

Step 1.4 Sector-material topics carried into the demand matrix

The cross-check in Step 1.3 is not left as commentary. The topics that are most sector-defining for seafood yet have no clean ESRS (and, equally, no VSME) home are carried into Phase 3 as a separate, explicitly flagged block of sector-material rows. After consolidation this block comprises *four* rows: supply-chain traceability (chain of custody); wild-stock status and sustainable wild sourcing; feed and marine-ingredient sourcing; and animal health and welfare. Antimicrobial use, treated as a fifth candidate in earlier drafting, is folded into the animal-health-and-welfare row rather than carried separately, for two reasons: it has no home of its own in GRI 13 (it is neither a topic nor a disclosure in the standard), and the certification schemes that demand it (ASC, BAP) treat antibiotic stewardship as an element of fish-health management. Wild-stock status partly overlaps ESRS E4 and is retained as a distinct sourcing-decision row; the coding rule that separates the two is pinned in Phase 3 (Section 4.5).

These rows are deliberately *not* derived from cohort consensus – the cohort does not report them at scale precisely because the ESRS taxonomy does not ask for them. They are admitted on a different, clearly labelled basis, and that basis differs by row. Two correspond to named GRI 13 topics: traceability to GRI 13.23 (Supply chain traceability) and animal health and welfare to GRI 13.11 (Animal health and welfare). The other two are not GRI 13 topics but are carried by the fishing/aquaculture *sector-specific disclosures* that GRI files under biodiversity – wild-stock under the "aquatic organisms caught or harvested" disclosure (13.3.11) and feed under the "aquaculture production" disclosure (13.3.10) – together with demand from EU sector legislation and major non-regulatory frameworks (the IUU and Fisheries Control Regulations, MSC/ASC chain of custody, GDST, FisheryProgress). Each gap row additionally carries a system tag (*wild*, *aqua*, or *both*) recording the production system to which it applies, so that a wild-capture-only or aquaculture-only firm excludes the rows that do not concern it.

Carrying these as a flagged block keeps the cohort-evidenced core and the sector-material additions epistemically distinct. **[→ RESULTS?]** In the matrix these rows score near-zero across the ESRS/VSME and investor columns and concentrate on the seafood-certification and regulatory columns; that contrast is an empirical result of the coding, not a premise, and may read better in the results chapter.

Final universe

The disclosure-topic universe used as the row dimension of the cross-framework demand matrix in Phase 3 comprises **32 rows**:

- 26 Tier 1 entries – DR-level cohort consensus ($\geq$5/6);
- 2 Tier 2 entries – topic-level cohort consensus ($\geq$5/6): E4 Biodiversity and S4 Consumer health and food safety;
- 4 sector-material entries – structural-gap seafood topics carried in on GRI 13 / sector-legislation grounds, flagged as additions rather than cohort-evidenced, and merging antimicrobial use into animal health and welfare.

Each entry carries metadata indicating its inclusion tier, its coding level (DR or Topic), a confidence flag distinguishing the 28 cohort-evidenced high-confidence entries from the 4 sector-material additions, and – for the gap rows – a production-system tag. The full funnel is therefore 70 ESRS topical DRs $\rightarrow$ 28 high-confidence entries (26 DR + 2 topic) $\rightarrow$ 32 rows after the four sector-material additions.

4.4 Phase 2 – Landscape mapping

Where Phase 1 fixes the *rows* of the demand matrix, Phase 2 fixes its *columns*: the instruments that make up the sustainability-disclosure ecosystem a European seafood-processing SME actually faces. The output is a structured catalogue – the Necessity × Domain map – recording, for every instrument, who demands it and what it's strategic importance is.

Organising frame. Instruments are arrayed on two axes. The *necessity* axis ranks each instrument by how imperative it is from the SME's perspective, in four tiers: Tier 1, legal licence to operate (binding law); Tier 2, market licence to operate (customer-demanded certifications and audits without which market access is lost); Tier 3, competitive expectation (reporting frameworks peers adopt); and Tier 4, strategic differentiation (voluntary leadership initiatives). The *domain* axis separates seafood-sector-specific instruments from general corporate-sustainability instruments. Two bands sit outside this grid: a Tier 0 food-integrity foundation – the food-safety and food-law floor every processor clears before sustainability strategy begins, in scope of the firm but out of scope of the disclosure analysis – and a Layer 2 enabling layer of methods, management systems and data infrastructure used to satisfy the demands above (the "methodology providers versus disclosure askers" distinction, Section 4.5).

Inclusion rule. An instrument is admitted to the Layer 1 grid only if it passes two gates, judged on the instrument's own text. *Gate 1 (ESG topic):* the instrument bears on an environmental, social or governance matter, which excludes fiscal, customs and data-protection law. *Gate 2 (own binding organisation-level obligation):* the instrument itself imposes on the firm a binding organisation-level obligation – a disclosure, due-diligence, certification, management-mechanism or market-access requirement – rather than an individual employee entitlement or an obligation that binds only Member States. Gate 2 admits process-and-mechanism duties (the OSH Framework Directive, the Whistleblowing Directive's internal-reporting-channel requirement) on the same footing as reporting duties, while excluding individual-entitlement labour law (the Work–Life Balance Directive) and Member-State-binding instruments (the Adequate Minimum Wages Directive). A third carve-out keeps product- and food-safety conformity in the Tier 0 foundation, even though the consumer-health topic re-enters Phase 3 through the frameworks that ask about it. The full candidate set – including instruments considered and rejected, with the gate each failed – is recorded in a scope log, so that membership is reproducible rather than a matter of recall.

Stakeholder source and obligation locus. Each admitted instrument carries two attributes that its Phase 3 column inherits. Its *source* records the dominant stakeholder whose demand the processor actually answers, determined by the test "does the instrument legally bind the processor?": where it does, the source is the regulator; where it does not, the source is the transmitting stakeholder – a customer audit, a capital-market expectation, or an industry/NGO initiative. For a sub-threshold SME selling B2B to large retailers, this rule reassigns several instruments whose nominal issuer is a regulator to the customer channel through which the SME actually experiences them: CSRD/ESRS, CSDDD and CDP are coded *customer*, because the dominant transmitter is the large customer's value-chain data request rather than the capital market, while the *investor* code is reserved for instruments whose primary intended user is capital (SFDR, the EU Taxonomy, IFRS S1/S2, and VSME via lenders). The *obligation locus* records separately whether the processor is the bound entity (*direct*), is reached through another party's duty (*indirect*), or merely operates within a regime that binds someone else (*context*).

Firm-profile conditionality. Several instruments apply only to firms with particular characteristics. These are flagged scope-conditional and governed in Phase 3 by firm-profile include-switches: the F-gas Regulation applies only where fluorinated refrigerants are used; the Pay Transparency Directive only at or above 150 employees; ISSF only to tuna processors; and RSPO only where palm is physically handled as an input. Setting the firm profile activates or deactivates these columns and excludes the inapplicable ones from that firm's demand sums. A small number of instruments are retained in the catalogue for ecosystem completeness but carry no scorable demand on the SME and are excluded from the demand sums: investor benchmarks that rate only large listed firms (the Coller FAIRR Protein Producer Index, the Dow Jones Sustainability Index), a keystone-company coalition the SME cannot join (SeaBOS), and a scheme-benchmarking body that imposes no direct obligation on the processor (GSSI).

Output. The result is the column dimension of the Phase 3 matrix: the full ecosystem catalogued in this phase – the [N ≈ 38] instruments carried into Phase 3, of which three investor and coalition benchmarks are retained but unscored – each tagged with its tier, domain, source, obligation locus, an independence weight (Section 4.6) and any firm-profile condition. The map is intentionally *not* mutually exclusive: a single topic reached by several instruments remains visible under each, because the cross-framework convergence that Phase 3 measures depends on that overlap being preserved rather than collapsed.

4.5 Phase 3 – Cross-framework demand mapping

The central analytical construct is a coverage matrix whose rows are the 32 disclosure topics of the final universe (Section 4.3) and whose columns are the ecosystem of frameworks, standards, certifications and initiatives catalogued in Phase 2. Rather than the provisional twelve-framework subset, the matrix codes the *full* Phase 2 landscape – approximately 37 instruments spanning the four stakeholder drivers (regulator, customer, investor, industry/NGO) – so that convergence is measured across the demand environment the SME actually faces. Each column carries, as attributes inherited from Phase 2, its primary stakeholder driver and its obligation locus (direct, indirect, context), and an independence weight used in the dependency-adjusted aggregation of Phase 4 (Section 4.6); columns that a given firm's profile renders inapplicable (e.g. a tuna-specific or refrigerant-specific instrument) are deactivated by firm-profile include-switches and excluded from that firm's sums.

The coverage scale. Each cell is coded on a three-level scale, defined ex ante:

Coverage scale (defined ex ante)
0 – Not covered: the framework does not address this topic, or its demand maps only to datapoints outside the row universe (see out-of-frame rule below).
1 – Touched: the framework references the topic but does not require reporting or certification on any of the topic's datapoints.
2 – Required: the framework explicitly requires the entity to report on, certify, or disclose at least one datapoint belonging to the topic.

Datapoint-grounded coding. A score is assigned against the *constituent datapoints* of a requirement, not against the title of its Disclosure Requirement. For the 28 cohort-evidenced rows, each DR is decomposed into its official EFRAG datapoints using the ESRS datapoint master (the EFRAG datapoint list, filtered to the same "keep" layer applied in Phase 1), producing a requirement-to-datapoint linkage; the 26 DR-level rows resolve to 356 datapoints and the two topic-level rows (E4, S4) to a further 152. A framework scores 2 on a requirement where its indicators demand at least one of that requirement's datapoints, 1 where it references the topic but maps to none of those datapoints as a reporting or certification requirement, and 0 where it is silent. This discipline makes each judgement concrete and reproducible: it converts "does framework f address topic r?" into "does f demand any datapoint of r?". **[→ RESULTS?]** It also surfaces corrections – for example, whistleblowing demand resolves to the governance row that actually carries whistleblowing datapoints rather than to an adjacent anti-corruption row – which, if reported, are findings about coding rather than method.

Coding levels. The granularity of each row matches the granularity at which cohort evidence supports it (the `Coding_Level` column). For the two topic-level rows the requirement comprises all datapoints across the topic's sub-DRs, and a framework is scored against the topic as a whole. For the four sector-material gap rows no ESRS datapoint exists by construction; these are scored instead against the criteria of the demanding frameworks themselves – for instance MSC Principle 1 stock-status reference points for wild-stock, GDST key data elements for traceability, and ASC antibiotic and survival criteria for animal health and welfare.

The wild-stock / E4 boundary. Because wild-stock status partly overlaps ESRS E4, a pinned rule prevents double-counting: stock-status and legal-sourcing demand (MSC Principle 1, the IUU catch-certificate regime, FisheryProgress) codes to the wild-stock row, whereas generic biodiversity and ecosystem-impact demand (ESRS E4, GRI 101, TNFD, and MSC Principle 2 on bycatch and habitat) codes to E4. A single framework may therefore legitimately score on both rows through different provisions.

Out-of-frame demand. A framework's indicator-level demand sometimes maps cleanly onto an ESRS datapoint that lies *outside* the cohort-defined universe. The Pay Transparency Directive's gender pay-gap requirement maps to ESRS S1-16, and the Working Time Directive to S1-11/S1-15 – none of which cleared the $\geq 5/6$ consensus and so have no row. Such demand is scored 0 against the in-universe rows; it is *not* reassigned to the nearest available row, since doing so would misattribute the demand and inflate that row. Instead it is recorded in a separate out-of-frame demand register (framework → demanded DR → reason out of frame). This preserves the cohort-grounded logic of the universe – a framework's contribution to the matrix is the *intersection* of its demand with the cohort-evidenced universe, not the union of everything it asks – while keeping the observation visible. **[→ RESULTS?]** That mandatory labour-standards disclosure (pay transparency, working time) and the voluntary sustainability-reporting universe occupy partly disjoint spaces is itself a finding for the discussion.

Methodology providers vs disclosure askers. The distinction between methodology providers (e.g. the GHG Protocol, ISO 14001, which supply accounting or process standards but do not themselves pressure an entity to report) and disclosure askers (e.g. CDP, SBTi, MSC, which require an entity to submit or certify information) is made explicit. Methodology-only frameworks are documented in the landscape mapping but held in the enabling layer and excluded from the demand sums.

Audit trail. Every non-zero cell is documented in a justification log with a one-line rationale, a citation to the source framework clause or indicator, *and* the identifier(s) of the ESRS datapoint(s) the demand maps to. This per-cell documentation constitutes the audit trail and protects the coding against reviewer challenge. Figure 1 sets out the decision applied to each cell.

4.6 Phase 4 - Hotspot analysis and stage assignment

Once the demand matrix is populated, several analytical operations are performed.

Aggregation by stakeholder type

The framework columns are collapsed into four stakeholder-driver buckets (regulator-SME, regulator-sector, customer, investor, industry/NGO), producing for each disclosure topic a count of frameworks demanding it within each stakeholder type.

Hotspot scoring

Two complementary measures are computed per topic:

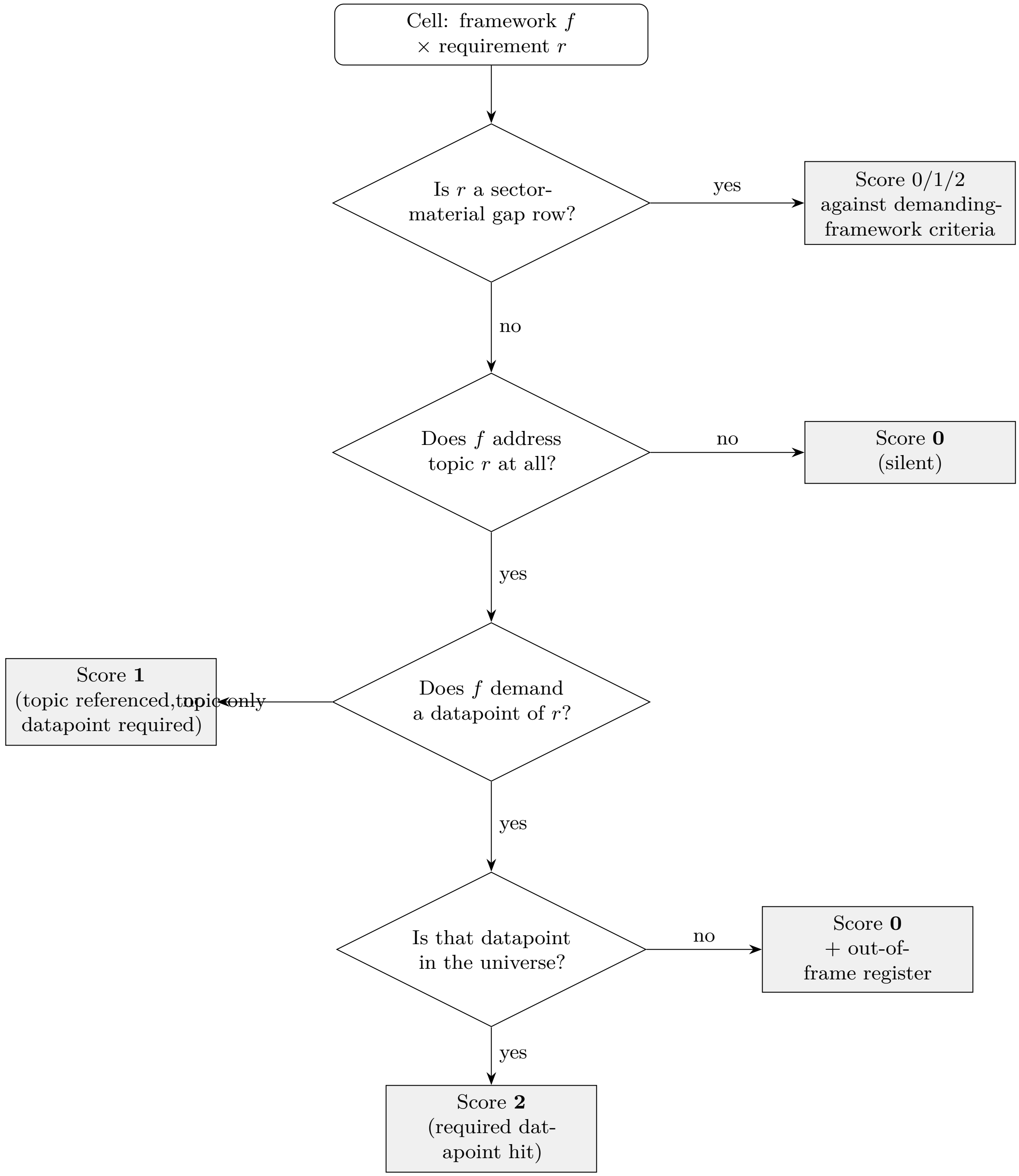

Figure 1: Per-cell coding protocol. Each framework × requirement cell is scored against the requirement's constituent ESRS datapoints; gap rows, which have no ESRS datapoints by construction, are scored against the demanding frameworks' own criteria; a framework silent on the requirement scores 0, and demand that maps only to out-of-universe datapoints also scores 0 and is logged separately.

- **Coverage breadth:** total number of priority frameworks with a non-zero coverage code for the topic.
- **Stakeholder diversity:** number of distinct stakeholder-driver types in which at least one framework requires the topic. This measure rewards cross-stakeholder convergence over within-stakeholder repetition.

Hotspot identification prioritises stakeholder diversity, on the conceptual argument that topics demanded by multiple stakeholder types deliver the highest leverage from a single capability investment.

Stage assignment

Topics are assigned to one of three development stages, with thresholds set after inspection of the empirical distribution:

- **Stage 1 - Foundational:** topics with $\geq$3 stakeholder types demanding and $\geq$6 frameworks covering. Multiple stakeholders independently identify these as material; reporting infrastructure built for one stakeholder serves several. Highest priority for SME initial investment.
- **Stage 2 - Expansion:** topics with 2 stakeholder types and 3–5 frameworks. Cross-stakeholder consensus is forming; build capability as resources allow.
- **Stage 3 - Specialist / regulatory-only:** topics covered by ESRS regulation or by one specialist framework only. Required for compliance where in scope, but not yet drawing broad demand.

The diagnostic tool

The diagnostic tool operationalises the stage-of-development logic for individual SMEs. It comprises two inputs and two outputs.

Input 1 - current reporting posture self-assessment: a short structured self-assessment (5–8 items) positioning the SME on the reactive-to-proactive spectrum. Indicative items: presence of formal sustainability strategy; presence of a dedicated sustainability function or role; whether reporting is initiated externally (reactive) or internally (proactive); presence of voluntary commitments (e.g., SBTi-aligned targets); whether disclosure data is reusable across multiple stakeholder requests.

Input 2 - stakeholder-pressure mapping: the SME identifies which stakeholders are actively requesting sustainability information (e.g., specific major retail customers and which questionnaire formats, specific certifications required for product placement, any investor pressure, any regulatory pressure beyond CSRD baseline). This input reflects the SME's actual external demand environment.

Output 1 - prioritised disclosure topics: the subset of Stage 1 topics whose demanding frameworks align with the SME's identified stakeholder pressures. These are the topics for which capability investment delivers maximum response coverage given the SME's specific situation.

Output 2 - staged roadmap: a recommended sequencing of capability investments across Stage 1 → Stage 2 → Stage 3, scaled to the SME's current posture and resourcing.

The diagnostic positioning on the reactive-to-proactive spectrum is operationalised through the self-assessment in Input 1, on the basis that reactive vs proactive is a behavioural rather than a measurable categorical state, and that the SME itself is best placed to characterise its current orientation.

4.7 Phase 5 - Case study validation

The framework is validated through one or two deep case studies of European mid-sized seafood processing SMEs (50–250 employees, primarily B2B suppliers to large EU retailers). For each case the following sequence is applied:

- Initial scoping interview (~60 min) to characterise the firm: size, operations, value-chain position, existing certification portfolio, current sustainability reporting practice, current customer/investor pressures.
- Application of the diagnostic tool — the firm completes the self-assessment and stakeholder-pressure mapping under researcher guidance.
- Generation of the recommended prioritised topic set and staged roadmap from the framework.
- Validation interview (~60 min) to discuss the recommendations: feasibility, perceived utility, gaps, misalignments with the firm's lived context. Findings inform refinement of the tool.

The case study sample is intentionally small (1–2 firms) and the validation is formative rather than confirmatory. Generalisation beyond the case is not claimed; the case studies are used to test usability and refine the diagnostic tool, not to validate the underlying empirical claims about the cohort or the cross-framework demand pattern.

4.8 Framework validation strategy

The seafood sustainability playing field (Section 3.4) is a conceptual framework, and the standards by which such a framework is assessed differ from those applied to a hypothesis. A conceptual framework is not proven; it is developed through a structured process and substantiated through convergent evidence (Jabareen, 2009; Meredith, 1993). The framework advances two distinct claims that require different forms of substantiation. The descriptive claim is that the seafood sustainability disclosure environment can be exhaustively and unambiguously organised by the necessity, domain and source axes. The explanatory claim, which is the more substantive and the more demanding to substantiate, is that capability corridors exist, such that capability investment compelled by a lower-tier mandatory requirement materially subsidises higher-tier strategic disclosure.

Five complementary validation strategies are employed; no single one is treated as conclusive, and the framework's credibility rests on their convergence.

Leg 1 - Theoretical grounding

Each axis of the framework is anchored in established theory rather than derived ad hoc. The necessity axis draws on legitimacy theory (Suchman, 1995) and institutional theory (DiMaggio & Powell, 1983); the source overlay on stakeholder-salience theory (Mitchell, Agle & Wood, 1997); the capability-corridor construct on the resource-based view (Barney, 1991) and the dynamic-capabilities literature (Teece, 2007); and the domain axis on materiality and sector-standard literature. Demonstrating that every structural component of the framework follows from prior theory substantiates its architecture as theoretically coherent.

Leg 2 - Descriptive completeness and inter-coder reliability

The descriptive claim is tested empirically during Phase 2. Every instrument identified in the landscape mapping is classified into exactly one necessity-by-domain cell with one primary source attribution. The framework is descriptively valid to the extent that this classification is exhaustive, in that no instrument requires a residual category, and unambiguous, in that instruments genuinely spanning cells are few and are documented as such. To establish that the classification is replicable rather than idiosyncratic, a second coder independently classifies a sample of instruments, and inter-coder agreement is reported using Cohen's kappa; agreement above the conventional threshold for substantial reliability substantiates that the framework can be applied consistently.

Leg 3 - Expert face validity

The framework is presented to expert informants, namely sustainability leads at the cohort companies and managers at the SME case-study firms, who are asked whether its tiers, domains and corridors correspond to their professional experience. This constitutes member checking, a recognised qualitative-validation technique (Lincoln & Guba, 1985). Expert feedback is documented systematically, including instances in which it challenges the framework, and revisions made in response to such feedback are reported as evidence of validation rather than concealed.

Leg 4 - Substantiation of the capability-corridor mechanism

The explanatory claim is substantiated through two converging methods. First, an input-overlap analysis codes the underlying data and process inputs required by each instrument situated on a corridor; demonstrating that, for example, MSC chain-of-custody, the GDST standard and ESRS value-chain disclosure draw on a common body of lot-level traceability data provides objective evidence that the corridor is real. Second, retrospective testimony is gathered from the large-reporter cohort: informants are asked directly whether capability built for a mandatory requirement reduced the marginal effort of subsequent disclosures. Convergence between the objective input-overlap evidence and the testimonial evidence substantiates the corridor mechanism more strongly than either method alone.

Leg 5 - Instrumental validation through case application

Finally, the framework is applied to the SME case studies described in Section 4.7. If it correctly locates a firm's current posture, correctly identifies the corridors offering that firm the greatest leverage given its stakeholder pressures, and is recognised by the firm as an accurate and actionable diagnosis, the framework is validated instrumentally, that is, by its demonstrated usefulness in application. This corresponds to the pattern-matching logic of case-study research (Yin, 2018): the framework predicts a pattern, and the case evidence is examined for correspondence to it.

Together, these five strategies validate the framework to a degree appropriate to the available evidence, and it is important to state the limits of this claim precisely. The descriptive claim, that the framework organises the observed ecosystem, can be substantiated robustly through Legs 1 to 3. The explanatory claim, that mandatory compliance subsidises strategic sustainability through capability corridors, is substantiated as plausible and evidenced through Leg 4 and as useful through Leg 5 but, given the small cohort and the limited number of case studies, is not claimed to be causally proven. The defensible position is that the evidence is consistent with the corridor mechanism, that firms report having experienced it, and that the instruments on a corridor demonstrably share inputs, rather than that a causal relationship has been established beyond doubt.

5. Expected outputs and chapter structure

The thesis produces five concrete artefacts mapping to chapter structure as follows.

Chapter	Central artefact	Form
Ch. 1 Introduction + Ch. 2 Conceptual framework	The reactive→proactive frame; the cross-stakeholder convergence argument	Prose; reactive vs proactive comparison table
Ch. 3 Methodology	The five-phase methodology, the 43-topic universe definition, the 0/1/2 coverage scale	Prose; methodology figure showing the funnel from 70 DRs → 36 → 43

Chapter	Central artefact	Form
Ch. 4 Cohort empirical analysis	Descriptive empirics of large-reporter disclosure practice	Heatmaps; per-topic and per-company coverage charts
Ch. 5 Cross-framework demand mapping	The central heatmap (33 topics $\times$ 37 frameworks); stakeholder aggregation	Master heatmap figure; per-stakeholder breakdown charts
Ch. 6 Stages-of-development roadmap + diagnostic tool	Stage 1/2/3 assignment per topic; the diagnostic tool architecture	Staged roadmap figure; diagnostic tool process diagram
Ch. 7 Case study validation	Application and refinement based on 1–2 SME case studies	Case study narratives with tool output
Ch. 8 Discussion + policy implications	Implications for SME sustainability practice; engagement with the EU simplification debate (Omnibus, VSME, sector standards)	Prose; brief policy recommendations
Appendix	Prioritised KPI shortlist for European seafood SMEs (Stage 1 + Stage 2 disclosures expanded to representative KPIs)	Structured table
Appendix	Full landscape mapping (37 frameworks with metadata)	Reference table
Appendix	Full coverage matrix (43 $\times$ 12 with per-cell justifications)	Audit table

6. Scope notes and known limitations

- **Empirical scope.** Findings apply to European mid-sized seafood processing SMEs (50–250 employees) primarily supplying large EU retailers under B2B arrangements. Extension to other SME profiles (smaller firms, primary producers, B2C operations) is treated as future work.
- **Cohort representativeness.** The five-company cohort represents an

upper bound of sector disclosure depth rather than an average. Inferences about typical practice are not made from this cohort; the cohort is used to establish what is empirically feasible to disclose, not what is typical.

- **Cross-framework coding subjectivity.** Coding of framework coverage on the 0/1/2 scale involves analyst judgement. Mitigation: ex-ante coding rubric, per-cell documentation with source citations, sensitivity analysis on threshold choices.
- **Case study generalisability.** One or two case studies cannot validate the framework empirically across the SME population. The case studies serve as usability tests of the diagnostic tool, not as confirmatory empirical evidence for the framework structure.
- **Static framework snapshot.** The regulatory and voluntary framework landscape is evolving rapidly (Omnibus simplification, VSME refinement, CSDDD phased entry, sector ESRS postponement). Findings reflect the landscape as observed in May 2026.

APPENDIX:

TOOL

What the SME sees - Step 1: Self-assessment (Input 1)

Where are you on the reactive ↔ proactive spectrum? (7 items, takes 5 minutes)

#	Question	SME's answer
1	Do you have a formal written sustainability strategy approved by leadership?	In development
2	Do you have a dedicated sustainability function or full-time role?	Part of another role (50%)
3	What primarily initiates your sustainability reporting work?	Mostly external requests, some internal
4	Do you have voluntary external commitments (e.g., SBTi, Net-Zero pledge)?	Considering — not yet committed
5	Is your sustainability data reusable across multiple customer requests?	Partially — rebuilt for each major customer
6	Do you measure and report on metrics that no customer or regulator currently requires?	No
7	Does sustainability factor into your 3-year strategic planning?	Limited — mentioned but not central

→ Score: 9 / 21 — early-stage reactive, transitioning.

Step 2: Active stakeholder pressures (Input 2)

Tick all that currently apply. Add detail where prompted.

Customer pressures (B2B retailer / buyer requirements):

- ✓ Sedex/SMETA audit required → *by Lidl, ICA*
- ✓ EcoVadis assessment requested → *by Albert Heijn*
- ✓ MSC chain-of-custody → *all wild-capture products*
- ✓ ASC certification (in progress) → *for farmed salmon launch*
- ✓ Customer ESG questionnaire (custom format) → *Lidl annual; ICA biannual*

☐ BRCGS / IFS Food (food safety) — *already certified, taken as given*

Investor / financial pressures:

- ✓ Bank sustainability-linked loan covenant → *small green loan from local bank, requires basic emissions reporting*

☐ CDP submission requested

☐ Equity investor ESG questions

Regulatory pressures (direct):

☐ CSRD direct (out of scope)

- ✓ VSME considering for next reporting year
- ✓ EU IUU regulation compliance (catch certificates for sourced wild fish)
- ✓ EU Control Regulation 2023/2842 — digital traceability rollout

Industry / NGO / Sector:

☐ GRI reporting

☐ Industry coalition / FAIRR engagement

☐ NGO scrutiny

Step 3: What the tool does (the logic)

The tool now cross-references:

- The SME's *active frameworks* (from Input 2): Sedex/SMETA, EcoVadis, MSC, ASC, Lidl-questionnaire, ICA-questionnaire, bank-loan-covenant, VSME, IUU, Control Reg
- The demand matrix (43 topics × 12 frameworks): which topics each of those active frameworks asks about
- Stage assignments per topic

Filters and ranks the Stage 1 topics by *how many of the SME's active frameworks demand each topic.*

Step 4: What the SME gets back — Output

Your situation: *Early-stage reactive (9/21). Active demand from 6 customer-driven frameworks, 1 investor-driven, 4 regulatory.*

Output 1 — Your top 10 priority topics (Stage 1, ranked by demand alignment with your pressures):

Rank	Topic	# of your active frameworks demanding this	Stakeholder types covered
1	Scope 1 + 2 GHG emissions	7 of your 11 (Sedex, EcoVadis, MSC chain-of-custody, Lidl Q, VSME, bank-loan, ...)	Customer, Investor, Regulator
2	Own workforce H&S incidents	6 (Sedex, EcoVadis, Lidl Q, ICA Q, ASC, VSME)	Customer, Regulator
3	Chain-of-custody / product traceability	6 (MSC, ASC, Lidl Q, IUU, Control Reg, ICA Q)	Customer, Regulator
4	Energy consumption	5 (EcoVadis, Lidl Q, VSME, bank-loan, ASC)	Customer, Investor, Regulator
5	Water use	4 (ASC, EcoVadis, Lidl Q, VSME)	Customer, Regulator
6	Value chain worker policies (forced labour, child labour)	4 (Sedex/SMETA, ICA Q, EcoVadis, Lidl Q)	Customer
7	Anti-corruption policy + incidents	4 (Sedex, EcoVadis, Lidl Q, VSME)	Customer, Regulator
8	Waste generated by processing	4 (Lidl Q, EcoVadis, VSME, ASC)	Customer, Regulator
9	Animal welfare (aquaculture-specific)	3 (ASC, BAP, NGO interest)	Customer
10	Antimicrobial use (aquaculture)	3 (ASC, BAP, F2F Strategy)	Customer, Regulator

Output 2 — Your staged roadmap (personalised):

Stage 1 — Foundation (start now, 6–12 months): Build capability on topics 1–5. These satisfy ≥4 of your active frameworks each. Recommended sequence: GHG (1) → traceability (3) → energy + water (4, 5) → H&S (2). Estimated coverage gain: ~70% of your current external requests answerable from one data infrastructure.

Stage 2 — Expansion (12–24 months, once Stage 1 is operational): Add topics 6–8 plus VSME basic module formalisation. This positions you to handle Lidl's deepening sustainability scorecard and EcoVadis level improvements, plus respond to additional customer ESG questionnaires without rebuild.

Stage 3 — Differentiation (24+ months, proactive moves): Topics 9–10 (animal welfare, antimicrobial use) — currently demanded by ASC + sector frameworks but not yet by most retailers. Investing here positions Nordic Whitefish as an aquaculture sustainability leader as you launch farmed salmon, ahead of likely retailer asks in 2027–28.

Positioning note:

Your self-assessment placed you at 9/21 — *early-stage reactive transitioning*. The Stage 1 priorities above, if implemented, would move you to *capability-driven proactive*, because your reporting infrastructure would be designed to serve multiple stakeholder demands simultaneously rather than rebuilt per request.

Ecosystem Glossary

APPENDIX

Glossary of the Seafood Sustainability Disclosure Ecosystem

Frameworks, regulations, certifications and standards — Layer 1 (demand) and Layer 2 (enablement).

Snapshot: June 2026. Regulatory references and statuses current as of this date.

Contents

How to use this glossary

This glossary catalogues every instrument in the seafood sustainability landscape mapped in this thesis. It works both as an orientation read: what each actor in the ecosystem actually is and who runs it, and as a reference appendix to the coverage matrix. Entries are grouped into two layers, mirroring the demand analysis in the methodology.

Layer 1 — Demand-side instruments (disclosure askers). Instruments that place a demand directly on the firm: they require it to report, certify, disclose or comply. These are the columns of the cross-framework demand matrix, sub-grouped by the stakeholder that drives them (EU horizontal, EU sector/operational, customer, investor, industry/NGO).

Layer 2 — Enablement / methodology layer. Instruments that provide the methods, accounting rules, benchmarks or data standards that make disclosure possible but do not themselves demand reporting from the firm. They are documented in the landscape yet held separate from the demand count — central to the capability-corridor argument rather than to the hotspot tally.

Dual-role instruments. A few sit on the boundary — TCFD/TNFD are frameworks that also shape investor asks; GDST is a data standard buyers may require; the EU Taxonomy informs financiers. Each is placed by its primary role, with the dual nature noted in its entry.

Each entry records, in a fixed metadata block: **the organisation behind the instrument**, its legal reference or founding year, its status, its stakeholder source and whether it is an asker or enabler, followed by its objective, its role in the seafood ecosystem, and — where relevant — an evidence note and an official source link.

Reading the flags. Following the project's evidence convention: A **Status flag** highlights instruments whose scope, timing or legal standing was actively changing as of June 2026 — chiefly the EU 'Omnibus' simplification (CSRD, CSDDD, ESRS), the postponed EUDR timetable, the voluntary status of the VSME, and the future application dates of the Forced Labour Regulation and EmpCo.

Layer 1 — Demand-side instruments (disclosure askers)

1A · EU horizontal regulation — disclosure, due diligence & finance

CSRD — Corporate Sustainability Reporting Directive

Behind it	EU (Commission proposal; co-legislated by Parliament & Council)
Reference / founded	Directive (EU) 2022/2464 — in force January 2023
Status · role	EU directive; mandatory, transposed nationally · Asker
Stakeholder source	Regulator (horizontal)
Official source	eur-lex.europa.eu — Directive (EU) 2022/2464

Objective. Expands and standardises mandatory corporate sustainability reporting across the EU, requiring in-scope undertakings to report under the ESRS on a double-materiality basis (impacts on people and planet, and sustainability matters financially material to the firm).

In the ecosystem. Sets the scope boundary that decides whether a seafood processor must report at all. Most 50–250-employee processors fall below its thresholds and are reached only as value-chain partners of larger in-scope customers.

Status flag. *Scope and timing are being narrowed by the 2025 'Omnibus I' simplification package (raising thresholds and delaying waves). Treat the in-scope population as a moving target and verify the consolidated text before citing thresholds.*

ESRS — European Sustainability Reporting Standards

Behind it	Developed by EFRAG (technical adviser); adopted by the European Commission
Reference / founded	Commission Delegated Regulation (EU) 2023/2772 — first set, 2023
Status · role	Delegated regulation under CSRD; mandatory for in-scope firms · Asker
Stakeholder source	Regulator (horizontal)
Official source	eur-lex.europa.eu — Del. Reg (EU) 2023/2772

Objective. The detailed disclosure standards that operationalise the CSRD: two cross-cutting standards (ESRS 1–2) plus ten topical standards (E1–E5, S1–S4, G1), all built on double materiality and a defined set of Disclosure Requirements (DRs).

In the ecosystem. The DR-level ruler against which this thesis maps all cohort and SME disclosure; it defines the topical Disclosure Requirements that the coverage matrix scores frameworks against.

Status flag. *A simplified ESRS set (fewer datapoints) is under EFRAG revision following the Omnibus mandate — the exact DR/datapoint counts may change.*

VSME — Voluntary Sustainability Reporting Standard for SMEs

Behind it	EFRAG (technical adviser to the European Commission)
Reference / founded	Delivered to the Commission Dec 2024; Commission Recommendation adopted 30 July 2025
Status · role	Voluntary EU standard; non-binding (not a delegated act, so no legal standing under CSRD Art. 29c yet) · Asker
Stakeholder source	Regulator (SME-proportionate floor)
Official source	ec.europa.eu — VSME Recommendation (30 July 2025)

Objective. A proportionate voluntary standard for SMEs outside CSRD scope, structured as a Basic Module and a Comprehensive Module, designed to cut the reporting load from roughly a thousand ESRS datapoints to around a hundred and to answer the most common value-chain and lender information requests.

In the ecosystem. The realistic regulatory reference point — and candidate 'home base' — for non-CSRD seafood SMEs; built specifically so a small supplier can respond to a large customer's ESRS-driven data request once rather than per buyer.

Status flag. *It is a Commission Recommendation, not a Delegated Act — adoption is encouraged but voluntary as of June 2026. Do not describe it as legally binding.*

CSDDD — Corporate Sustainability Due Diligence Directive

Behind it	EU
Reference / founded	Directive (EU) 2024/1760
Status · role	EU directive; phased transposition · Asker
Stakeholder source	Regulator (horizontal)
Official source	eur-lex.europa.eu — Directive (EU) 2024/1760

Objective. Requires large companies to conduct human-rights and environmental due diligence across their chains of activities — identifying, preventing, mitigating and accounting for adverse impacts. It is a conduct/due-diligence directive rather than a reporting standard.

In the ecosystem. Drives value-chain due-diligence expectations (e.g. forced labour in fishing fleets) that cascade contractually onto SME suppliers even when those SMEs are not themselves in scope.

Status flag. *Scope, timing and the civil-liability regime are being scaled back under the 2025 Omnibus package; the first-wave application date has moved. Verify the consolidated text.*

EU Taxonomy — Regulation on the establishment of a framework to facilitate sustainable investment

Behind it	EU
Reference / founded	Regulation (EU) 2020/852
Status · role	EU regulation; directly applicable · Asker / Enabler
Stakeholder source	Regulator (capital-market facing)
Official source	eur-lex.europa.eu — Regulation (EU) 2020/852

Objective. A classification system defining environmentally sustainable economic activities against six environmental objectives, using technical screening criteria, 'do-no-significant-harm' tests and minimum social safeguards.

In the ecosystem. Shapes how banks and investors assess 'green' exposure; reaches seafood SMEs mainly through lender and investor alignment questions rather than as a direct obligation.

SFDR — Sustainable Finance Disclosure Regulation

Behind it	EU

Reference / founded	Regulation (EU) 2019/2088 — applies from 2021
Status · role	EU regulation; targets financial market participants · Asker
Stakeholder source	Regulator (investor-facing)
Official source	eur-lex.europa.eu — Regulation (EU) 2019/2088

Objective. Requires financial market participants and advisers to disclose how they integrate sustainability risks and to report principal adverse impacts (PAI) of their investments.

In the ecosystem. Indirect pressure on the firm: SFDR-bound funds and banks pass PAI data requests down to investee and borrower seafood companies.

1B · EU sector & operational regulation — seafood, food & labour

CFP — Common Fisheries Policy

Behind it	EU
Reference / founded	Regulation (EU) No 1380/2013
Status · role	EU regulation; directly applicable · Asker
Stakeholder source	Regulator (sector)
Official source	eur-lex.europa.eu — Reg (EU) 1380/2013

Objective. The overarching EU framework for managing fishing fleets and conserving stocks — fishing-opportunity setting (TACs/quotas), fleet-capacity ceilings, the landing obligation and a maximum-sustainable-yield (MSY) exploitation objective.

In the ecosystem. Tier-1 legal licence to operate for wild-capture sourcing; it binds Member States, and processor-facing obligations flow downstream through the Control Regulation, IUU and CMO rather than from the CFP itself.

IUU Regulation — Illegal, Unreported and Unregulated Fishing Regulation

Behind it	EU
Reference / founded	Council Regulation (EC) No 1005/2008 — applies from 2010
Status · role	EU regulation; directly applicable · Asker
Stakeholder source	Regulator (sector)
Official source	eur-lex.europa.eu — Reg (EC) 1005/2008

Objective. Prevents, deters and eliminates illegal, unreported and unregulated fishing, chiefly through a catch-certificate scheme: imported wild-capture products must carry a catch certificate validated by the flag state.

In the ecosystem. Anchor of the external traceability corridor — a validated catch certificate is a Tier-1 precondition for placing imported wild fish on the EU market, complementary to (not a subset of) the internal Control Regulation.

Control Regulation (recast) — Fisheries Control Regulation

Behind it	EU
Reference / founded	Regulation (EU) 2023/2842, amending Reg (EC) 1224/2009
Status · role	EU regulation; directly applicable (phased) · Asker
Stakeholder source	Regulator (sector)
Official source	eur-lex.europa.eu — Reg (EU) 2023/2842

Objective. Modernises EU fisheries control and enforcement — including full-chain digital traceability, mandatory electronic catch documentation and tighter monitoring of the EU fleet and the EU market.

In the ecosystem. Governs internal EU-fleet control and EU-market traceability (the sibling to IUU's external import scheme); pushes lot-level digital traceability that feeds the traceability-capability corridor.

CMO Regulation — Common Market Organisation for fishery & aquaculture products

Behind it	EU
Reference / founded	Regulation (EU) No 1379/2013
Status · role	EU regulation; directly applicable · Asker
Stakeholder source	Regulator (sector)
Official source	eur-lex.europa.eu — Reg (EU) 1379/2013

Objective. Organises the EU market for fishery and aquaculture products, including common marketing standards and mandatory consumer information — commercial and scientific species name, production method, catch/production area and gear category.

In the ecosystem. Binds the processor at the point of sale; its mandatory origin and method labelling overlap directly with chain-of-custody and consumer-information disclosures.

Fisheries Labour Directive — Implementation of the ILO Work in Fishing Convention (C188)

Behind it	EU (implementing social-partner agreement on ILO C188)
Reference / founded	Council Directive (EU) 2017/159
Status · role	EU directive · Asker
Stakeholder source	Regulator (sector, social)
Official source	eur-lex.europa.eu — Dir (EU) 2017/159

Objective. Sets minimum working and living conditions for fishers aboard vessels (contracts, hours/rest, repatriation, accommodation, medical care).

In the ecosystem. Binds vessel owners rather than the processor; reaches the firm as value-chain content via buyer SMETA audits and ESRS S2, and as a baseline for forced-labour scrutiny.

EUDR — EU Deforestation Regulation

Behind it	EU
Reference / founded	Regulation (EU) 2023/1115 — phased application
Status · role	EU regulation; directly applicable · Asker
Stakeholder source	Regulator (sector-adjacent)
Official source	eur-lex.europa.eu — Reg (EU) 2023/1115

Objective. Prohibits placing on (or exporting from) the EU market certain commodities linked to deforestation — cattle, soy, palm oil, wood, cocoa, coffee, rubber — without due-diligence statements backed by geolocation evidence.

In the ecosystem. Relevant to aquaculture feed sourcing (soy and palm derivatives) and to paper/board packaging, and to natural-ecosystem-conversion topics.

Status flag. *Application was postponed by 12 months: large/medium operators from 30 Dec 2025 and micro/small operators from 30 June 2026 — confirm the current date for your size class.*

PPWR — Packaging and Packaging Waste Regulation

Behind it	EU
Reference / founded	Regulation (EU) 2025/40 — in force 11 Feb 2025

Status · role	EU regulation; main obligations apply from 12 Aug 2026 · Asker
Stakeholder source	Regulator (horizontal)
Official source	eur-lex.europa.eu — Reg (EU) 2025/40

Objective. Replaces the 1994 Packaging Directive with directly-applicable rules on packaging minimisation, recyclability, recycled-content targets and labelling for all packaging placed on the EU market.

In the ecosystem. Touches resource-use and waste (E5) plus the packaging data that retailers increasingly request; binds the processor as the entity placing packaged product on the market.

EmpCo — Empowering Consumers for the Green Transition Directive

Behind it	EU
Reference / founded	Directive (EU) 2024/825 — in force 26 Mar 2024
Status · role	EU directive; transpose by 27 Mar 2026, apply from 27 Sep 2026 · Asker
Stakeholder source	Regulator (horizontal)
Official source	eur-lex.europa.eu — Dir (EU) 2024/825

Objective. Amends the Unfair Commercial Practices and Consumer Rights Directives to ban generic, unsubstantiated green claims ('eco-friendly', 'climate neutral') and to regulate sustainability labels.

In the ecosystem. Constrains 'sustainable' and eco claims on seafood packaging and raises the evidentiary bar for certification-based marketing.

Status flag. *It is the consumer-law half of the EU greenwashing agenda; the companion Green Claims Directive (substantiation rules) was still in negotiation as of mid-2026 — do not conflate the two.*

Forced Labour Regulation — Prohibition of products made with forced labour on the Union market

Behind it	EU
Reference / founded	Regulation (EU) 2024/3015 — published Dec 2024
Status · role	EU regulation; applies from 14 Dec 2027 · Asker
Stakeholder source	Regulator (horizontal)

Official source	eur-lex.europa.eu — Reg (EU) 2024/3015

Objective. Prohibits placing or making available on the EU market — or exporting — any product made with forced labour at any stage of its production, with investigation and enforcement powers for authorities.

In the ecosystem. Adds enforcement teeth to forced-labour risk in fishing and processing supply chains; documented due diligence reduces investigation exposure and reinforces S2 value-chain work.

Status flag. *Applies from 14 Dec 2027 — it is a future obligation, not yet enforceable. State the date when citing it.*

General Food Law — General principles and requirements of food law

Behind it	EU (establishes EFSA and the RASFF alert system)
Reference / founded	Regulation (EC) No 178/2002
Status · role	EU regulation; directly applicable · Asker
Stakeholder source	Regulator (food safety)
Official source	eur-lex.europa.eu — Reg (EC) 178/2002

Objective. Lays down general food-safety principles, the 'one-step-back / one-step-forward' traceability duty, and the Rapid Alert System for Food and Feed (RASFF).

In the ecosystem. Tier-1 food-integrity baseline; underpins food-safety incident, recall and traceability topics that sit beneath both regulatory and customer (GFSI) demands.

OSH Framework Directive — Safety and health of workers at work

Behind it	EU
Reference / founded	Directive 89/391/EEC
Status · role	EU directive; transposed nationally · Asker
Stakeholder source	Regulator (horizontal, social)
Official source	eur-lex.europa.eu — Dir 89/391/EEC

Objective. Establishes the employer's general duty to ensure workers' health and safety — risk assessment, prevention principles, information and training.

In the ecosystem. Legal baseline behind own-workforce health-and-safety (S1) disclosure; especially salient for cutting lines, machinery and cold stores in seafood processing.

Working Time Directive — Organisation of working time

Behind it	EU
Reference / founded	Directive 2003/88/EC
Status · role	EU directive; transposed nationally · Asker
Stakeholder source	Regulator (horizontal, social)
Official source	eur-lex.europa.eu — Dir 2003/88/EC

Objective. Sets minimum requirements for working hours, rest periods, breaks, paid annual leave and night work.

In the ecosystem. Underpins the working-conditions elements of own-workforce (S1) disclosure.

VMP Regulation — Veterinary Medicinal Products Regulation

Behind it	EU
Reference / founded	Regulation (EU) 2019/6 — applies from 2022
Status · role	EU regulation; directly applicable · Asker
Stakeholder source	Regulator (sector)
Official source	eur-lex.europa.eu — Reg (EU) 2019/6

Objective. Governs authorisation and use of veterinary medicines, including measures to curb antimicrobial use and resistance (e.g. restrictions on prophylactic and metaphylactic antibiotics).

In the ecosystem. The regulatory anchor for the aquaculture antimicrobial-use (AMU) topic, alongside the Farm to Fork antimicrobial-reduction target — one of the five structural-gap topics in the matrix.

1C · Customer-driven standards & audits

MSC — Marine Stewardship Council

Behind it	Independent non-profit (founded 1997 by Unilever & WWF; fully independent from 1999), London
Reference / founded	Founded 1997

Status · role	Voluntary certification — Fisheries Standard + Chain of Custody · Asker
Stakeholder source	Customer
Official source	msc.org

Objective. Certifies wild-capture fisheries against a science-based sustainability standard (stock status, ecosystem impact, management) and tracks certified product through the supply chain via a Chain-of-Custody standard, assessed by accredited third-party conformity-assessment bodies.

In the ecosystem. A de-facto market licence for many wild-capture retail listings; its Chain of Custody sits squarely on the traceability corridor. Wild-caught scope only.

Evidence. Peer-reviewed work links certification to healthier stock indicators — Gutiérrez et al. (2012, PLoS ONE) found the ecolabel conveys reliable stock-health information — though effect-size and selection-bias debates persist (see 'Shifting focus: the impacts of sustainable seafood certification', PLoS ONE). Use as associative, not causal, evidence.

ASC — Aquaculture Stewardship Council

Behind it	Non-profit (founded 2010 by WWF & IDH), Utrecht
Reference / founded	Founded 2010
Status · role	Voluntary certification — species standards + Chain of Custody · Asker
Stakeholder source	Customer
Official source	asc-aqua.org

Objective. Certifies responsibly farmed seafood against species-specific environmental and social standards (water quality, feed, disease/antimicrobials, labour), with a chain-of-custody linked to the MSC CoC programme.

In the ecosystem. Market expectation for farmed seafood such as salmon, shrimp and pangasius; covers welfare, antimicrobials, water and feed. Aquaculture scope only.

Evidence. ASC and MSC merged their chain-of-custody programme; for farmed-stock outcome evidence the literature is thinner and more recent than for wild MSC — flag any effectiveness claim as emerging. [LIKELY]

BAP — Best Aquaculture Practices

Behind it	Global Seafood Alliance (formerly Global Aquaculture Alliance), USA
Reference / founded	From 2002 onwards
Status · role	Voluntary certification — multi-star, full-chain · Asker
Stakeholder source	Customer
Official source	bapcertification.org

Objective. Certifies aquaculture facilities across the chain — hatchery, farm, feed mill and processing plant — against environmental, social, food-safety and animal-welfare criteria, using a 'star' rating for chain completeness.

In the ecosystem. Alternative or complement to ASC and the only major scheme certifying feed mills directly; strong in North American buyer requirements. Aquaculture scope.

GLOBALG.A.P. — Good Aquaculture Practice (Aquaculture standard)

Behind it	FoodPLUS GmbH (origin EUREPGAP, 1997), Germany
Reference / founded	1997 (EUREPGAP) / aquaculture module 2007
Status · role	Voluntary B2B certification · Asker
Stakeholder source	Customer
Official source	globalgap.org

Objective. Business-to-business good-practice certification covering food safety, environmental management and worker welfare at the production stage, with an aquaculture-specific module.

In the ecosystem. A common retailer prerequisite, particularly for farmed produce; often the production-stage layer beneath a retailer's social and food-safety asks. Aquaculture scope (for the module in view).

Friend of the Sea — Friend of the Sea

Behind it	World Sustainability Organization (founded 2008), Italy
Reference / founded	Founded 2008
Status · role	Voluntary certification (wild + farmed) · Asker
Stakeholder source	Customer
Official source	friendofthesea.org

Objective. Certifies wild and farmed seafood (and other products) against sustainability criteria, with an ecolabel.

In the ecosystem. An additional ecolabel option with lower retail penetration than MSC/ASC but present in the landscape; covers both production systems.

IFS Food — International Featured Standards — Food

Behind it	IFS Management GmbH (German/French/Italian retail origin)
Reference / founded	Founded 2003
Status · role	Voluntary, GFSI-benchmarked food-safety certification · Asker
Stakeholder source	Customer
Official source	ifs-certification.com

Objective. Certifies food-safety and quality-management systems for processors supplying (mainly continental-European) retail.

In the ecosystem. Tier-2 market licence — effectively non-negotiable for retail listing; its supplier-control requirements anchor the supplier-engagement corridor.

BRCGS — Brand Reputation Compliance Global Standards

Behind it	Owned by LGC (formerly British Retail Consortium)
Reference / founded	From 1998
Status · role	Voluntary, GFSI-benchmarked food-safety certification · Asker
Stakeholder source	Customer
Official source	brcgs.com

Objective. Global food-safety and quality certification widely required by UK and international retailers.

In the ecosystem. A near-substitute for IFS Food with the same market-licence function; buyer choice between them is largely geographic.

FSSC 22000 — Food Safety System Certification 22000

Behind it	Foundation FSSC

Reference / founded	Founded 2009 (built on ISO 22000)
Status · role	Voluntary, GFSI-benchmarked · Asker
Stakeholder source	Customer
Official source	fssc.com

Objective. Food-safety system certification combining ISO 22000 with sector-specific prerequisite programmes.

In the ecosystem. The third common GFSI-benchmarked option; interchangeable with IFS/BRCGS in most buyer specifications.

Sedex / SMETA — Supplier Ethical Data Exchange / Sedex Members Ethical Trade Audit

Behind it	Sedex — membership platform; SMETA is its audit methodology, UK
Reference / founded	Founded 2004
Status · role	Voluntary social-audit platform / methodology · Asker
Stakeholder source	Customer
Official source	sedex.com

Objective. Shares ethical-trade data among members and provides a standard social-audit methodology (SMETA) covering labour, health & safety, environment and business ethics.

In the ecosystem. A common retailer-required social audit; feeds S1/S2 worker and H&S topics and sits on the supplier-engagement corridor.

EcoVadis — EcoVadis

Behind it	EcoVadis SAS — private ratings company (founded 2007), France
Reference / founded	Founded 2007
Status · role	Voluntary sustainability rating / assessment · Asker
Stakeholder source	Customer
Official source	ecovadis.com

Objective. Scores suppliers on environment, labour & human rights, ethics and sustainable procurement through a documentation-based assessment yielding a medal/percentile scorecard.

In the ecosystem. Frequently requested by large retail and B2B buyers; its broad multi-topic scope makes it a strong hotspot driver in the demand matrix.

1D · Investor-driven frameworks

CDP — CDP (formerly Carbon Disclosure Project)

Behind it	CDP — non-profit, UK/global
Reference / founded	Founded 2000
Status · role	Voluntary disclosure system (investor- and buyer-requested) · Asker
Stakeholder source	Investor
Official source	cdp.net

Objective. Runs a global environmental disclosure system (climate, water security, forests) through scored questionnaires (A–D) used by investors and large buyers to request standardised data.

In the ecosystem. A common investor and large-customer climate & water questionnaire; strong overlap with GHG and water topics, and a frequent conduit by which buyer demand reaches the SME.

SBTi — Science Based Targets initiative

Behind it	Founded by CDP, UN Global Compact, WRI and WWF (now an incorporated entity)
Reference / founded	Founded 2015
Status · role	Voluntary target-validation · Asker
Stakeholder source	Investor
Official source	sciencebasedtargets.org

Objective. Defines and independently validates corporate emissions-reduction targets aligned with climate science (1.5°C pathways), including FLAG (Forest, Land and Agriculture) guidance relevant to feed and protein.

In the ecosystem. Signals a proactive climate posture; validated targets feed CDP and ESRS E1 climate disclosure.

TCFD — Task Force on Climate-related Financial Disclosures

Behind it	Established by the Financial Stability Board; monitoring transferred to the ISSB in 2023
Reference / founded	Created 2015; recommendations 2017
Status · role	Voluntary recommendations — now largely absorbed into IFRS S2 · Asker / Enabler
Stakeholder source	Investor
Official source	fsb-tcfd.org

Objective. Provided a four-pillar framework — governance, strategy, risk management, and metrics & targets — for climate-related financial disclosure.

In the ecosystem. The conceptual backbone now carried forward by IFRS S2 and ESRS E1; the standalone task force was disbanded after 2023, so treat it as legacy architecture rather than a live ask.

IFRS S1 / S2 — ISSB Sustainability & Climate Disclosure Standards

Behind it	International Sustainability Standards Board, under the IFRS Foundation (created at COP26, 2021)
Reference / founded	Standards issued 2023
Status · role	Voluntary baseline; becoming mandatory via national adoption · Asker
Stakeholder source	Investor
Official source	ifrs.org — ISSB

Objective. A global baseline of investor-focused disclosure: S1 (general sustainability-related financial information) and S2 (climate), on a financial-materiality ('outside-in') basis.

In the ecosystem. The capital-market counterpart to ESRS; most relevant where a seafood firm has an internationally listed parent or investors in adopting jurisdictions.

SASB Standards — Sustainability Accounting Standards Board Standards

Behind it	Now under the ISSB / IFRS Foundation (via the 2022 Value Reporting Foundation merger)
Reference / founded	Codified 2018
Status · role	Voluntary industry-specific metrics · Asker / Enabler

Stakeholder source	Investor
Official source	sasb.ifrs.org

Objective. Industry-specific, financially-material sustainability metrics, including an agriculture/protein lens.

In the ecosystem. Feeds investor metric expectations and now underpins the industry guidance within the ISSB standards rather than operating as a separate ask.

Coller FAIRR — Coller FAIRR Protein Producer Index

Behind it	FAIRR Initiative — investor network founded by Jeremy Coller (2015)
Reference / founded	Index since 2018
Status · role	Voluntary investor benchmark · Asker
Stakeholder source	Investor
Official source	fairr.org

Objective. Ranks the largest listed animal-protein producers (including aquaculture) on ESG risk factors — feed, antibiotics, welfare, emissions, deforestation — for investor use.

In the ecosystem. Sector-specific investor scrutiny that a firm is rated by rather than a member of; directly relevant for large/listed producers, largely out of scope for an unlisted 50–250-employee SME.

DJSI — Dow Jones Sustainability Indices

Behind it	S&P Global (Corporate Sustainability Assessment)
Reference / founded	Since 1999
Status · role	Voluntary investor index / assessment · Asker
Stakeholder source	Investor
Official source	spglobal.com — CSA

Objective. Selects index constituents via an annual Corporate Sustainability Assessment scoring ESG performance.

In the ecosystem. Relevant mainly to large listed parents; signals investor-grade ESG performance and is, like FAIRR, a rating-of rather than a membership.

TNFD — Taskforce on Nature-related Financial Disclosures

Behind it	Market-led, government-backed taskforce (launched 2021)
Reference / founded	Recommendations finalised 2023
Status · role	Voluntary recommendations · Asker / Enabler
Stakeholder source	Investor
Official source	tnfd.global

Objective. A framework (the LEAP approach) for assessing and disclosing nature- and biodiversity-related dependencies, impacts, risks and opportunities.

In the ecosystem. The emerging biodiversity/ecosystem lens, highly relevant to marine impacts and dependencies (E4); an EFRAG–TNFD correspondence mapping links it to ESRS.

1E · Industry / NGO voluntary frameworks

GRI — GRI Standards (incl. GRI 13)

Behind it	Global Reporting Initiative — independent standards body (founded 1997), Amsterdam
Reference / founded	GRI 13 'Agriculture, Aquaculture and Fishing Sectors', 2022
Status · role	Voluntary reporting standards · Asker
Stakeholder source	Industry / NGO
Official source	globalreporting.org

Objective. The most widely used global sustainability-reporting standards, on an impact ('inside-out') materiality basis; GRI 13 enumerates the topics likely material to the agriculture, aquaculture and fishing sector.

In the ecosystem. The sector-materiality benchmark used in this thesis's topic-universe cross-check and the single most seafood-relevant existing disclosure map; an EFRAG–GRI interoperability index (Nov 2024) aligns GRI with ESRS.

UN Global Compact — United Nations Global Compact

Behind it	United Nations
Reference / founded	Founded 2000
Status · role	Voluntary corporate-citizenship initiative · Asker
Stakeholder source	Industry / NGO

Official source	unglobalcompact.org

Objective. Commits signatories to ten principles on human rights, labour, environment and anti-corruption, with an annual Communication on Progress.

In the ecosystem. A common voluntary commitment and normative reference for governance and social topics; entry-level credibility signal rather than a data standard.

Move to Minus 15 — Move to -15 °C Coalition

Behind it	Industry coalition (convened with DP World and supply-chain partners)
Reference / founded	Initiated 2023
Status · role	Voluntary initiative · Asker
Stakeholder source	Industry / NGO
Official source	movetominus15.com

Objective. A coalition exploring raising the frozen-food storage standard from -18 °C to -15 °C to cut energy use and emissions without compromising safety.

In the ecosystem. A sector-decarbonisation initiative touching energy (E1) and cold-chain operations; directly relevant to frozen-seafood logistics.

Status flag. *Governance and the evidence base are still developing; treat membership/standard details as [ASSUMPTION] and verify before citing.*

ISSF — International Seafood Sustainability Foundation

Behind it	Science-and-industry partnership (scientists, tuna processors, WWF)
Reference / founded	Founded 2009
Status · role	Voluntary (tuna-focused) · Asker
Stakeholder source	Industry / NGO
Official source	iss-foundation.org

Objective. Promotes science-based tuna conservation and vessel/processor compliance measures, including the ProActive Vessel Register and audited conservation measures.

In the ecosystem. Tuna-specific; relevant only to canned-tuna processors. Wild-caught scope.

FIPs / FisheryProgress — Fishery Improvement Projects / FisheryProgress.org

Behind it	FisheryProgress.org operated by FishChoice in partnership with the Sustainable Fisheries Partnership (SFP)
Reference / founded	Platform from ~2017
Status · role	Voluntary improvement mechanism (not a certification) · Asker
Stakeholder source	Industry / NGO
Official source	fisheryprogress.org

Objective. Time-bound, multi-stakeholder projects that move a fishery toward sustainability, with public progress tracking on a central platform; many are engineered toward an eventual MSC assessment.

In the ecosystem. A sourcing pathway for not-yet-certified fisheries and buyer-recognised improvement evidence — but because FIP requirements largely derive from MSC, it is treated as a non-independent signal (collapsed with MSC) in the convergence count, while its lower strength is reflected by a reduced independence weight. Wild scope.

Evidence. FIP effectiveness is contested in the literature: reviews flag stalled progress and weak accountability once market access is gained ('Shifting focus: the impacts of sustainable seafood certification', PLoS ONE). Present FIP participation as intent, not outcome.

SeaBOS — Seafood Business for Ocean Stewardship

Behind it	Science–business initiative linked to the Stockholm Resilience Centre
Reference / founded	Founded 2016
Status · role	Voluntary CEO-level coalition · Asker
Stakeholder source	Industry / NGO
Official source	seabos.org

Objective. Connects the largest ('keystone') seafood companies with science to drive ocean stewardship on IUU fishing, traceability and climate.

In the ecosystem. A large-player, agenda-setting initiative rather than an SME-facing demand; useful as a direction-of-travel reference, aspirational for a 50–250-employee firm.

RSPO — Roundtable on Sustainable Palm Oil

Behind it	RSPO — multi-stakeholder non-profit, Malaysia/Switzerland
Reference / founded	Founded 2004
Status · role	Voluntary certification + supply-chain models · Asker
Stakeholder source	Industry / NGO
Official source	rspo.org

Objective. Certifies sustainable palm oil along the supply chain via production standards and chain-of-custody models.

In the ecosystem. Relevant where palm derivatives appear in aquaculture feed or processing inputs; intersects with the EUDR due-diligence obligation.

Layer 2 — Enablement / methodology layer

These instruments underpin the demand-side layer. They supply accounting methods, management-system processes, benchmarks or data formats. A firm builds capability on them, but they do not themselves request a disclosure — which is why they are excluded from the demand count yet central to the capability-corridor argument.

GHG Protocol — Greenhouse Gas Protocol

Behind it	World Resources Institute (WRI) & WBCSD
Reference / founded	Corporate Standard, 2001
Status · role	Accounting methodology (not certifiable) · Enabler
Stakeholder source	Cross-cutting
Official source	ghgprotocol.org

Objective. The dominant corporate GHG accounting and reporting methodology, defining Scopes 1, 2 and 3 and the rules for boundary-setting and calculation.

In the ecosystem. Underlies every climate disclosure (ESRS E1, CDP, SBTi); it does not itself demand reporting but makes emissions data comparable — a pure capability enabler.

ISO 14001 — Environmental Management Systems

Behind it	International Organization for Standardization
Reference / founded	1996 / 2015 revision
Status · role	Management-system standard (certifiable) · Enabler

Stakeholder source	Cross-cutting
Official source	iso.org — ISO 14001

Objective. Specifies requirements for an environmental management system (EMS) on a plan-do-check-act basis.

In the ecosystem. Process backbone for environmental data; supports E-topic capability without itself being a disclosure asker.

ISO 45001 — Occupational Health & Safety Management Systems

Behind it	International Organization for Standardization
Reference / founded	2018
Status · role	Management-system standard (certifiable) · Enabler
Stakeholder source	Cross-cutting
Official source	iso.org — ISO 45001

Objective. Specifies requirements for an occupational health & safety management system.

In the ecosystem. Underpins own-workforce H&S (S1) data and processes; the management-system counterpart to the OSH Directive's legal duty.

ISO 50001 — Energy Management Systems

Behind it	International Organization for Standardization
Reference / founded	2011 / 2018 revision
Status · role	Management-system standard (certifiable) · Enabler
Stakeholder source	Cross-cutting
Official source	iso.org — ISO 50001

Objective. Specifies requirements for an energy management system, including energy-performance monitoring.

In the ecosystem. Supports the quality and auditability of energy-consumption data (ESRS E1).

GSSI — Global Sustainable Seafood Initiative

Behind it	GSSI — public-private partnership
Reference / founded	Founded 2015
Status · role	Benchmarking tool (meta-standard) · Enabler
Stakeholder source	Customer (meta)
Official source	ourgssi.org

Objective. Benchmarks seafood certification schemes against FAO-based criteria so buyers can recognise equivalence between recognised schemes.

In the ecosystem. Reduces certification fragmentation — it tells buyers which schemes (MSC, ASC, BAP, GLOBALG.A.P.) it recognises rather than demanding data itself; this is why it is treated as a meta-layer and collapsed, not counted as an independent demand.

GFSI — Global Food Safety Initiative

Behind it	The Consumer Goods Forum
Reference / founded	Founded 2000
Status · role	Benchmarking tool (meta-standard) · Enabler
Stakeholder source	Customer (meta)
Official source	mygfsi.com

Objective. Benchmarks food-safety schemes (IFS, BRCGS, FSSC 22000) for mutual recognition — 'certified once, accepted everywhere'.

In the ecosystem. The reason IFS, BRCGS and FSSC behave as near-substitutes: buyers accept any GFSI-benchmarked scheme, so they should not be triple-counted as independent demands.

GDST — Global Dialogue on Seafood Traceability

Behind it	GDST — non-profit foundation (industry-led, 60+ partners)
Reference / founded	Convened 2017; GDST 1.0 standard, 2020
Status · role	Voluntary data / interoperability standard · Enabler
Stakeholder source	Industry / NGO
Official source	thegdst.org

Objective. Defines the key data elements and interoperability formats for digital seafood traceability across wild and farmed supply chains.

In the ecosystem. The technical spine of the traceability corridor: it lets traceability data flow across IUU/Control-Reg requirements, MSC/ASC chain-of-custody and ESRS value-chain reporting. Buyers may require it, giving it a dual asker/enabler character.

UN SDGs — UN Sustainable Development Goals

Behind it	United Nations (Agenda 2030)
Reference / founded	2015
Status · role	Normative goal framework (not a reporting standard) · Enabler
Stakeholder source	Cross-cutting
Official source	sdgs.un.org

Objective. Seventeen goals and 169 targets providing a shared global agenda for sustainable development; SDG 14 ('Life below water') is the most directly seafood-relevant.

In the ecosystem. Used by firms to frame and communicate contribution; a reference frame rather than a data demand, which is why it sits in the enabler layer.

Selected literature & evidence base

Sources cited in the evidence notes above, plus key peer-reviewed work on seafood-certification effectiveness. URLs marked **(verified)** were confirmed live during research; entries marked [**verify citation**] should be checked for exact authorship, year and pagination before inclusion in the bibliography.

1. Gutiérrez, N. L. et al. (2012). "Eco-Label Conveys Reliable Information on Fish Stock Health to Seafood Consumers." PLoS ONE 7(8): e43765. PMC3424161 — *(verified) Evidence that MSC-certified stocks trend healthier; associative, not causal.*

2. "Shifting focus: the impacts of sustainable seafood certification." PLoS ONE (open access). PMC7239462 — *(verified URL) Reviews certification outcomes and FIP accountability concerns. [verify citation] for authors/year.*

3. Sampson, G. S. et al. (2015). "Secure sustainable seafood from developing countries." Science 348(6234): 504–506. — *[verify citation] Widely cited on FIPs and market access for developing-country fisheries.*

4. EFRAG (2024). GRI–ESRS Interoperability Index; EFRAG–TNFD correspondence mapping (2024). — *Official interoperability mappings underpinning the framework-connection logic. Available via efrag.org.*

5. FAO. Guidelines for the Ecolabelling of Fish and Fishery Products from Marine Capture Fisheries (basis for GSSI benchmarking). — *[verify citation] Primary normative reference behind GSSI recognition criteria.*

Compiled June 2026. All official-source links and regulatory statuses were current at the snapshot date; EU sustainability law (CSRD, CSDDD, ESRS, EUDR) was under active simplification and should be re-checked against EUR-Lex before citation.